

Research Note: The Physics of Coexistence

— Force, Charge, Mass, and Time in Satta—

Unit Interaction

Author: Raghava Mohan Madhwapathi (analyticmadhyasthdarshan.org)

Edited on: August 5, 2026, 12:20 AM IST

Status: Internal research note (not a catalog entry). Compiled to support drafting on *The Ontology of Coexistence* §1.2 and related studies.

Scope: Every explicit statement located in the primary texts (SB, MVD, JV, AVD, JVD, KD) that uses physics vocabulary — pressure, motion, force, magnetism, electricity, heat, charge/excitation (*āvesh* — context-dependent, see §8), mass, weight, gravitation, atomic structure (nucleus, orbits, elements, and the nature of the constituent particle/*ansh* itself), states of matter, density, radioactivity, indestructibility/conservation, cosmogony, time — to describe either (a) how *satta* (Omnipotence/Omnipresence) and units (*ikai*) relate, or (b) how units relate to one another. Citations use the project’s tag system (SB = *Samadhanatmak Bhautikvad* / “Resolution Centred Materialism”; MVD = *Madhyasth Darshan Coexistentialism*; JV = *Jeevan Vidya: An Introduction*; AVD = *Adhyatmavad*; JVD = *Janvad*; KD = *Manav Karm Darshan*, Hindi-only — KD quotes are our working translations; full working translation in `References/Madhyasth-Darshan/KD-Karm-Darshan-English/KD-Karm-Darshan-English.md`, see Method note) with page numbers verified directly against the source PDFs (see Method note at the end).

Structure. Every numbered section and subsection below is split into two clearly separated parts: **What MD says** — a numbered list of paraphrased claims taken from the primary texts, each with a page citation — and **Our reading** — this note’s own interpretation, cross-references, and flagged tensions. Treat the numbered claims as the raw material to re-verify against the source PDFs before quoting in a catalog Study (via `Scripts/_quote_tool.py verify`); treat “Our reading” as commentary, not text. §14 (Synthesis) and §15 (Open Tensions) are, by their nature, entirely “Our reading” — there is no separate claims list for them.

KD’s stated core principle (*siddhant*, title page) is **śram–gati–pariṇām** — effort–motion–result — the same triple SB p. 62 unfolds from saturation (§4). Its section 3 (chapters 3.1–3.15) is the most physics-dense stretch anywhere in the corpus: atomic structure (ch. 3.1, p. 53), development in the atom (3.2, p. 59), heat and the Earth’s balance (3.7, p. 72), state–motion (3.8, p. 83), quantity (3.9, p. 87), force–power (3.11, p. 101), reflection–projection (3.12, p. 109), pressure/flow/wave/electromagnetic force (3.13, p. 117), place–direction–distance–area–angle (3.14, p. 124), and time (3.15, p. 133).

1. The base claim: two poles, one coexistence

Everything below sits on top of one recurring assertion: existence has exactly two aspects — formless, actionless *satta* and formful, active *prakriti* (nature, as countless units) — and neither produces the other.

What MD says:

1. Consciousness (Omnipotence, *satta*) and matter are inseparably present; all of existence is, in essence, nature (matter) saturated in Omnipotence (consciousness) — neither is ever observed arising from the other (SB, p. 48).
2. Omnipotence itself is formless and omnipresent; nature within it is endowed with form, properties, essential nature, and dharma, and is inseparable from it — there is no place where Omnipotence is absent (SB, p. 48).
3. Nature, saturated in Omnipotence, exists as countless discrete units, each one surrounded, submerged, and soaked in it (SB, p. 48).

Our reading: The relation-word the texts use for this bond is *samprikt* — saturated: soaked, submerged, surrounded. It is not “created by,” “emitted from,” or “caused by.” Whatever physical-sounding vocabulary follows (force, energy, charge, weight) has to be read against this base claim, because the texts are explicit that *satta* itself never acts.

2. Satta is explicitly denied motion, pressure, and change

This is the texts’ own starting point for physical vocabulary — not an inference the study is making, but a direct statement, repeated in at least three of the five sources.

What MD says:

1. Omnipotence remains free of motion and pressure, while the whole of nature existing within it is present with motion and pressure — pressure being the compulsion toward attraction and repulsion arising under environmental (mutual) influence (SB, p. 49).
2. Consciousness/knowledge is omnipresent, permeative, transparent, and free of activity (devoid of wave or pressure); being changeless, it is not the root cause of the universe’s origin — coexistence is that root cause, and all change occurs only within nature as saturated in consciousness (MVD, p. 74).
3. Formless existence, as absolute energy, is eternally present without motion, wave, or pressure (SB, p. 50).

Our reading: pressure, attraction, and repulsion — the vocabulary of physical force — are placed explicitly at the level of unit-to-unit mutuality (“arising under environmental influence”), never as something *satta* does to a unit. This is the texts’ own firewall between the ground and physical activity.

2.1 The same firewall applied to cosmogony: the Big Bang is rejected by name. JV contains the corpus’s one direct engagement with a named theory of modern cosmology,

and it deploys exactly the §2 doctrine against it.

What MD says:

1. Omnipotence (Space) has no attribute by which it could apply a force — there are no waves, motion, or pressure in Space — so a hypothesis that the universe began with a Big Bang whose resulting force set everything in motion does not hold; every entity is instead energised, active, and inspired directly within Omnipotence (JV, p. 149).
2. Units are distinct from one another, and yet able to join with one another, because Omnipotence exists in between them; this is the basis for composition and decomposition of entities in nature (JV, p. 149).

Our reading: the Big Bang is not disputed on observational grounds (redshift, background radiation, etc. go unmentioned) but ruled out *a priori*: any cosmogonic first-force would have to be a force applied by the ground itself, and the ground is definitionally forceless. Note this sits on the same pages as the zero-attraction passage (§10.1) — in the source they are one continuous argument. Claim 2 also restates the “distance is satta” claim of SB p. 79 (§4) from the other direction: satta-in-between is simultaneously what keeps units distinct *and* what lets them join. This pairs with §11 below (no creation, no annihilation): the darshan’s alternative to cosmogony is not a different origin story but the denial that an origin event is needed at all — coexistence is *anaadi*, beginningless.

3. Two kinds of energy: Absolute vs. Relative

MVD names this distinction directly, and it is probably the single most useful primitive for organising everything else in this note.

What MD says:

1. Relative energy (*sapeksh oorja*) is the power that does not become apparent without the mutuality of units — pressure, waves, and fields among physical/chemical entities, and also heat, sound, and electricity, are all named as relative energy (MVD, p. 40).
2. Absolute and relative energy are mutually necessary: without absolute energy there is no basic impulsion in matter, and without matter absolute energy is not revealed; nature is eternally present in absolute energy (MVD, p. 40).
3. The entire material universe — its state and its motion alike — is saturated within and contained by absolute energy (MVD, p. 45).

Our reading: *Absolute energy* = *satta* itself, as the inherent, ever-present ground — inseparable from matter but not a force acting on it. *Relative energy* = everything that shows up only *between* units: pressure, waves, fields, heat, sound, electricity. Claim 1 is the clearest single statement in the corpus for “which physical phenomena are unit–unit, and which are satta–unit”: by MVD’s own definition, **every named physical force (heat, sound, electricity, pressure, waves, fields) is relative energy — i.e., strictly inter-unit.** Satta’s

own energy is never itself named as one of the physical forces; it is what the physical forces are various manifestations of, at one remove.

4. The saturation bond itself is described in force language

Here is the tension worth sitting with. Section 2 above establishes that *satta* does not act, has no pressure, no wave, no motion. But when the texts describe what saturation *does* to a unit, they switch into force vocabulary — repeatedly, not as a one-off metaphor.

What MD says:

1. Units' mutual recognition and regulation arise solely from their saturation and energisation in Omnipotence; this submersion is why every unit has space all around it — “being soaked itself is forcefulness” (SB, p. 57).
2. The force in a unit is essentially magnetic force, signifying its being soaked in Omnipotence; because this forcefulness runs through the unit constantly, the constancy of effort is self-evident (SB, p. 61).
3. Saturation in uniform energy is itself forcefulness; forcefulness is itself basic impulsion; basic impulsion is itself activity; activity is itself effort-motion-result; and effort-motion-result is itself development and its continuity (SB, p. 62).
4. Every existent, as a consequence of its saturation in Omnipotence, is endowed with forcefulness manifested as magnetic forcefulness — forcefulness is existentially inherent in every atomic particle (SB, p. 98).
5. Forcefulness is expressed because formful existence remains saturated in formless existence; every unit — molecules, atoms, even the particles contained within atoms — remains regulated within a definite, fixed distance from every other, and that distance is itself formless existence (SB, p. 79).
6. Every entity, from the smallest to the biggest unit, is submerged, soaked, and surrounded in the omnipresent; it is by virtue of being energy-endowed that force-endowment, and magnetic-force-endowment, is understood — it is this force by which the particles of an atom recognise one another in mutuality and participate in order (KD, p. 54; our translation).
7. Whatever we measure as distance, amid mutuality, we are measuring only the pervasive entity itself — every distance measurement is a measurement of *satta* (KD, pp. 125–126).

Our reading: the texts' own resolution (implicit, not stated as a resolution) is that saturation is an *eternal endowment*, not an *event* — “forcefulness” names what is always-already true of a soaked unit, not a transmission that happens at a moment in time, the way one billiard ball strikes another. Claim 3's chain (saturation → forcefulness → basic impulsion → activity → effort-motion-result) is presented as one continuous unfolding “internal” to the bond, not a causal relay between separate things. Even so, the literal choice of “magnetic

force” (चुम्बकीय बल) for a bond that is also called actionless is a real interpretive knot — see §15.

Claim 6 grounds recognition-and-regulation (the two effects claim 1 derives from saturation) in the same endowment — so the SB and KD chains match link for link: saturation → energy-endowment → (magnetic) force-endowment → recognition → regulation/order.

Claim 5 (SB p. 79) is also the clearest textual basis for “distance is satta”: the gap between any two particles or units is not empty in the sense of *nothing there* — it is formless existence itself. Space (a name the texts also give *satta* — see MVD p. 40’s list: Uniform Energy, Space, Knowledge, Consciousness, Absolute Energy) is not a container *satta* sits inside; it is *satta*, seen from the side of separation between bounded things. Claim 7 (KD ch. 3.14) makes the same claim operational, extending it to measurement itself — the one place the unmeasurable ground is said to be what rulers in fact measure.

5. The Five Forces table — physics mapped onto *jeevan*’s faculties

This is the most striking single artifact in the corpus for this question: SB contains an actual table pairing the four fundamental forces of physics, plus a fifth, against the five faculties of *jeevan* (SB, p. 85):

Insentient Powers — The Five Forces	<i>Jeevan</i> faculty	Sentient Power
Electromagnetic Force	<i>Mun</i>	Hope
Gravitational Force	<i>Vritti</i>	Thought
The Weak Interaction Force	<i>Chitta</i>	Visualisation
The Strong Interaction Force	<i>Buddhi</i>	Resolve
Mediative Force	<i>Atma</i>	Realisation

What MD says:

1. It is only in the atom that all five forces are seen to be fully operative; in post-forms — molecules and molecular compositions — the mediative force is no longer apparent, which is why the study and teaching of all five forces is accessible only through the atom (SB, p. 84).
2. Mediative activity is the manifestation of mediative force and power, situated at the centre of the atom, continuously controlling the generative and degenerative forces; insentient powers are subject to erosion, whereas sentient powers remain inexhaustible (SB, p. 84).

Our reading: the darshan takes over the standard four-force inventory of physics wholesale (electromagnetic, gravitational, weak, strong) and adds one the texts consider missing — a “mediative force” (*madhyastha bal*), located at the atomic nucleus, whose job is to regulate/normalise the other four (“generative and degenerative powers”). This fifth force is presented as *the* reason an insentient atom can, on developing, become a sentient one — mediative force is insentient nature’s structural analogue of *atma*. It is worth being careful in the paper about what kind of claim this is: it is not proposing a fifth force for physics to detect: it is a structural analogy (insentient force : atom :: sentient faculty : jeevan), used to argue that jeevan’s five-faculty architecture is not arbitrary but mirrors a five-fold structure the darshan already claims to see in insentient matter. Whether this is meant descriptively (physics literally has five forces once you count correctly) or analogically (a template shared across orders) is not disambiguated in the passage itself — flag as open in §15.

6. Atomic structure: nucleus, orbits, and the absence of a numbered periodic table

This is the most direct material for “atomic number and orbital shells,” and it turns out to be more developed than the Five Forces table alone suggests. The darshan gives every atom — insentient or sentient — the same generic architecture: a central particle-cluster (nucleus, *madhyansh*) surrounded by particles arranged in concentric orbits (*parivesh*, literally “surroundings,” translated here as orbit/shell).

What MD says:

1. Particles in every atom are situated either in the nucleus or in the surrounding orbits; the first orbit’s one or more particles continuously revolve around the nucleus, and the same holds for the second, third, fourth, and further orbits; space pervades all around the atom and around the particles within its constitution (MVD, p. 78).
2. The kind, state, and existential measure of atoms are determined by differences in the number of particles in their nucleus and their orbiting particles (MVD, p. 42).
3. The basic element in the constitution of all large entities is the atom, and the constitution principle of atoms of all elements is the same — a nucleus and orbiting particles. All orders of nature arise from the activeness of particles in atoms and their position in the development progression (MVD, p. 149).

Our reading: the darshan does have a real “shell model” — nucleus plus a variable number of concentric orbits, with particle count in each determining the atom’s “kind” (*jaati*), its overall size/mass, and (via distance from the nucleus) its charge state. Claim 3 states this architecture is **one universal principle across all “elements” and all four orders** — material, bio, animal, knowledge-order atoms are not built from different kinds of parts; they differ only in how many particles occupy which orbits and how “active” (in the effort/ascending sense) each orbit is.

6.1 Growth rule: how orbits and mass build up together. AVD gives the clearest statement connecting orbital structure to weight (cross-reference §10).

What MD says:

1. The atom's constitution is a nucleus at the centre plus dependent particles at a well-defined distance, orbiting the nucleus — motion-paths (orbits) are an integral part of the atom. More particles in an atom means more orbits too; as the number of particles in orbits increases, the number of particles in the nucleus increases as well — this is how weight-bondage appears in atoms (AVD, p. 185).

Our reading: growth is monotonic and coupled — more orbital particles requires a larger nucleus, and that pairing is explicitly what weight is (§10). This is the qualitative version of what KD sharpens into an inequality (§6.4a).

6.2 Orbital distance sets charge/regulation. AVD ties the same architecture to §8's charge discussion (the $\pm$ sense — see §8's terminology note), giving the nucleus an explicit regulatory role over the orbits.

What MD says:

1. Regulation of an atom's generative/degenerative excess is the grandeur of its central particle: an increase or decrease in the precise distance of orbital particles from the nucleus is what “generative” or “degenerative” names, and this is kept in equilibrium by the strength and power of the nucleus (AVD, p. 103).

Our reading: this is the same distance-from-nucleus definition of generative/degenerative excitation that reappears independently in §8 — a case of AVD and SB converging on the same mechanism without cross-reference.

6.3 Differential activity of orbits, order by order. MVD states that which orbits are “active” differs systematically by order of nature — this is as close as the texts come to something like an electron-configuration rule.

What MD says:

1. Only the particles in the last orbit of all kinds of atoms within the material order are active through effort; only the outermost and second-to-last orbits of insentient atoms are actively engaged in compounding activities; particles in the fourth, third, and second orbits in the sentient atom (animal order) are partially active; in the knowledge order, jeevan expresses four-and-a-half activities, with activeness in the fourth, third, and second orbits (MVD, pp. 78–79).

Our reading: this is the darshan's nearest analogue to an electron-configuration/valence-shell rule — which orbit is “doing the work” changes systematically as one moves up through the four orders of nature, without the underlying nucleus-plus-orbits architecture changing at all (§6, claim 3).

6.4 The one case with a fully named orbital model: jeevan itself. When the “atom” in question is a constitutionally complete, sentient one, the nucleus and its four orbits get

specific names and a fixed activity count — the clearest “shell diagram” anywhere in the corpus.

What MD says:

1. The nucleus of the sentient unit (a constitutionally complete atom) is called *atma*; the particles of its first orbit are called *buddhi*, its second orbit *chitta*, its third orbit *vritti*, and its fourth orbit *mun* (MVD, p. 78).
2. *Atma*, as jeevan’s nucleus, has exactly two activities — realisation (*anubhav*) and authenticity (*pramanikta*). *Buddhi*’s first orbit has two activities — enlightenment (*bodh*) and resolve (*sankalp*). *Chitta*’s second orbit — contemplation (*chintan*) and visualisation (*chitran*). *Vritti*’s third orbit — deliberation (*tulan*) and analysis (*vishleshan*). *Mun*’s fourth orbit — taste (*asvadan*) and selection (*chayan*). Together these are jeevan’s ten activities (JV, p. 92).

Our reading: the only place the orbit count is fixed at a specific number is *jeevan* — exactly four orbits plus a nucleus, ten activities total, each orbit given a specific psychological/cognitive name rather than a numeric label. Ordinary (insentient) atoms are explicitly allowed more orbits (“first, second, third, fourth, and so on,” §6 claim 1) — the texts do not cap or enumerate how many.

6.4a KD’s quantitative additions: minimum atom, nucleus-to-orbit ratio, the four-orbit equilibrium, and the 108 species. KD ch. 3.1 (*Paramanu Sanrachna evam Anu Rachna*, pp. 53–57) restates the §6 architecture and then goes quantitatively further than SB/MVD/AVD anywhere (our translations throughout).

What MD says:

1. A minimum atom requires at least one nucleus particle and at least one orbiting particle — with less than this, an atom does not obtain (KD, p. 53).
2. Whatever number of particles are working in the orbits, at least that many are present in the nucleus — the nucleus count may be greater, never fewer (KD, p. 56).
3. Orbits number one or more; when orbits exceed four, the possibility of decreasing remains; when fewer than four, the possibility of increasing remains — always (KD, p. 56).
4. Difference in the number of working particles produces difference in conduct; conduct-difference is the basis on which the species (*prajaati*) of atoms, and their number, are distinguished — humans have so far claimed 108 species of atoms on this Earth, and remain hopeful of finding more (KD, p. 55).
5. Weight depends on the number of nucleus particles (*madhyansh*) inherent in an atom; by virtue of this inherent weight, atoms recognise one another and aggregate — this is called a molecule (KD, p. 56).

Our reading: claim 1 is a minimum-constitution axiom (1 nucleus particle + 1 orbital particle); claim 2 is a nucleus ≥ orbits particle-count inequality, sharper than AVD p. 185’s qualitative growth rule (§6.1); claim 3 makes four orbits an equilibrium/attractor of the orbit

count — bearing directly on the jeevan-four-orbits question flagged in §15; claim 4 is the corpus’s one acknowledgement of the chemical-element inventory (see the revised “what is not there” note under §6.5); claim 5 keys weight specifically to the *nucleus* count, not the total count — a refinement of §10.3.

6.4b KD on particle transfer: excitation precedes displacement. KD p. 57 adds a process account absent from the other five texts — how atoms change their particle count at all.

What MD says:

1. Change in the number of particles, and the consequence of that change, is *pariṇām* (result); the activity itself is called *prasthāpan–visthāpan* (absorption–displacement). Before any particle is displaced from an atom, that atom is found to be excited (*āveshit*) — it is only as a result of being excited that displacement becomes possible. The displaced particle merges into some other atom (absorption); afterward, the displaced atom settles into natural-state motion, and the absorbing atom, having absorbed the particle, is likewise established in natural-state motion (KD, p. 57).
2. In every atom, activity takes the form of effort, motion, result: effort is the grandeur of the existent state, motion is the grandeur of spreading effect, and result is the grandeur of a different existent state (KD, p. 117).

Our reading: this is the corpus’s transmutation mechanism, and it knits §6, §8, and §9.2 into one loop: excitation (*āvesh* in its state sense, §8) is the *precondition* for particle emission; emitted particles are absorbed by other atoms; both donor and receiver then relax into natural-state motion; and since particle count fixes species (§6) and *pariṇām* names exactly this count-change, “result” — the third term of KD’s own śram–gati–pariṇām principle — turns out to mean, at the material level, atomic transmutation. Radioactivity (§9.2’s excited, overfull atoms shedding internally) is then a special case of *visthāpan*.

6.5 Above the atom: states of matter (solid, liquid, gaseous) and the elemental/compound/mixture taxonomy. The texts also carry a small but explicit bulk-matter taxonomy sitting on top of the atomic model.

What MD says:

1. All matter exists in solid, liquid, or gaseous (air) form, and can be found in elemental, compound, or mixture states (MVD, pp. 41–42).
2. An elemental state (*tatvik*) is the composition of a homogeneous group of atoms (MVD, p. 42).
3. Physical entities exist in solid and gaseous forms only, while chemical substances manifest in solid, liquid, and gaseous states (SB, p. 97).
4. Chemical substances show their splendour in solid, liquid, and gaseous forms — for instance, water as ice, liquid water, and steam (SB, p. 142).
5. Once solid substances take on their complete form, the grandeur of chemical substances emerges through solid, liquid, and gaseous processes; the formation of biological cells

and prana sutra is one historic milestone from the solidification of chemical entities (SB, p. 145).

6. The insentient world — material order and biological order — exists in solid, liquid, and gaseous states (JV, p. 14).

Our reading: three states, not four — plasma never appears as a state of matter anywhere in the corpus (the word occurs once in SB, p. 204, as a translation of *rasa* in the seven bodily *dhatus*). Claim 2's "elemental = a homogeneous group of atoms" is the closest the corpus comes to defining a chemical element, and it is consistent with §6's picture: an element is a *population* of same-kind atoms (same particle count), not an entry in a numbered table. One quirk worth preserving: claim 3 gives *physical* entities only solid and gaseous forms, reserving the full solid–liquid–gas triple for *chemical* substances — liquidity is apparently treated as evidence of chemical (*rasa*-yielding) activity, which coheres with claim 5's claim that biological cells emerge from solidification processes of chemical entities.

What is not there: none of the six texts assign a specific particle count to a specific named element (hydrogen, oxygen, iron, gold, etc. all appear only as generic illustrations of "-ness," e.g. "ironness," never with a stated number of nucleus/orbit particles) — so there is no periodic table proper, no per-element atomic-number assignments, and no electron/proton/neutron vocabulary anywhere in this corpus (verified for KD as well: no Devanagari loan-words for electron, proton, or neutron occur). §6.4a claim 4 does, however, explicitly acknowledge the element inventory *as a human claim*, so the darshan is aware of, and accepts the reality behind, a discrete numbered series of atomic kinds; what it declines to reproduce is the table's contents. What exists instead is the growth law (more particles → more orbits → bigger nucleus → more weight → different "kind"), now sharpened by KD's constraints (§6.4a: minimum 1+1 constitution; nucleus ≥ orbit count; four-orbit equilibrium) — still without the per-element numbers that would let you look up carbon's structure the way you would in a chemistry textbook.

6.6 The particle itself: functional role by position, not particle kind. The nucleus/orbit split above is also a functional split at the particle level, not merely a positional one.

What MD says:

1. The nucleus particle(s) (*madhyansh*) carry out mediative activity and are more potent than the other particles (MVD, p. 174).
2. The orbital/dependent particles (*āshrit ansh*) carry out generative-degenerative (*sam-viṣam*) activity (MVD, p. 174; see §8).
3. In the one fully-named case — *jeevan* — *atma* (the nucleus) is named directly as the sentient unit's most developed *ansh* (MVD, p. 317; cf. §6.4).

Our reading: the darshan differentiates *ansh* by role and, in *jeevan*'s case, by a gradient of subtlety (the outermost orbit, *mun*, being the most sense-bound, and *atma* at the centre be-

ing the most refined) — never by kind. Nucleus vs. orbit is a division of function (mediative vs. generative-degenerative), not of substance.

6.7 Every *ansh* is a “unit,” already saturated in Omnipotence.

What MD says:

1. Particles carry the capacity for activeness because they, like any other unit, are saturated in Omnipotence (MVD, p. 89).
2. It is by virtue of being energy-endowed, and thereby (magnetic-)force-endowed, that an atom’s particles are able to recognise one another in mutuality at all (KD, p. 54; cf. §4, claim 6).

Our reading: §4 establishes that saturation-forcefulness is a standing endowment of every unit in existence; claims 1–2 extend this explicitly to the particle level. An *ansh*’s activeness is therefore not conferred by the atom it joins — it precedes and enables that joining.

6.8 The isolated particle: restless and incomplete outside a constitution.

What MD says:

1. A particle that remains separate within the omnipresent substance fails to evidence the orderliness that is its natural intent; it therefore persists in a whirling motion, in search of another particle, until it is absorbed into an existing atom or joins another particle to form a new one (KD, p. 56).
2. Only on absorption does the receiving and donating atom settle into natural-state motion (KD, p. 57; §6.4b).

Our reading: a lone *ansh* is not pictured as a stable, self-sufficient little sphere. Its natural condition in isolation is restlessness and incompleteness — motion without settlement — and this restlessness is itself what drives combination. This is the same restlessness-until-completion logic that KD’s “hungry”/“undigested”/“satisfied” vocabulary (§6.4b) applies at the atom level, already present in miniature at the level of a single particle — combination-seeking is not imposed on an *ansh* from outside; it is what the particle’s own incompleteness already drives it toward.

6.9 Inside the atom: “own existence” at a fixed distance — and the limits of what is attributed to a lone particle.

What MD says:

1. Every particle within an atom clarifies its own existence (*apna astitva*) by holding a definite, fixed distance from the other particles in their mutuality — the collective discipline of this fixed spacing constitutes the atom’s boundary and motion-path (SB, ch. 4, “Existence and Development of Atom in Existence,” approx. pp. 60–61; page pin approximate, pending PDF verification, unlike this note’s other SB citations).
2. The gap between any two particles is not empty but is formless existence itself (SB, p. 79; §4, claim 5).

Our reading: two further limits are worth making explicit, neither of which the texts state as a single claim but both of which follow from material already cited: **(a)** sentience/insentience is a status of the completed atom, never of an isolated particle — when an atom crosses into constitutional completeness, every particle it comprises participates in the transition equally, with no suggestion that some particles are “already” sentient ahead of others; **(b)** no intrinsic, standalone, measurable property (shape, size, a charge value) is ever assigned to a particle considered alone — weight, charge, and an atom’s hungry/undigested/satisfied condition (§6.4b) are all explicitly relational, a function of an *ansh*’s position and count relative to the others in its constitution (§§8, 10), consistent with the corpus’s general “relative energy” doctrine (§3) that named physical quantities are apparent only in mutuality. A lone *ansh*, by the texts’ own logic, is restless and incomplete (§6.8) but not further quantifiable until it is in relation to at least one other particle.

Our reading, pulled together (§6.6–6.9): the *ansh* is the same “unit” concept as everything else in the darshan’s ontology, recursively applied at the smallest scale the corpus names — saturated, force-endowed, individually existent — but it behaves less like a classical-mechanics body with fixed, particle-intrinsic properties than like a relational-ontology part: almost everything sayable about it is sayable only once it is in relation to another *ansh*.

7. Motion: rotational, orbital, vibrational — and “excited” (SB: “charged”) vs. “natural-state” motion

What MD says:

1. Every unit exhibits motion in rotational, orbital, and vibrational form; as an atom develops, its vibrational motion keeps increasing, and its stability increases alongside it — the moment constitutional completeness is achieved, the atom’s vibrational motion reaches its natural ceiling (SB, pp. 53, 59).
2. Motion, in every unit, is categorised as rotational, orbital, or vibrational; this motion, together with environmental pressure, results in displacement — “place” meaning equal energy, or space and energy-fullness, since every unit is a bounded body from six directions. Every unit exists, in its mutuality, either in natural-state motion or in charged (excited) motion, the primary cause of charged motion being pressure from the combination of naturalness and environment (SB, p. 59).

Our reading: the darshan’s basic kinematics is: every unit (atomic or otherwise) always has rotational + orbital + vibrational motion, intrinsically (this is closer to “zero-point” activity than to Newtonian rest). What varies is whether that motion stays in its *natural state* (*swabhav gati*) or becomes *excited motion* (*aveshit gati*) under pressure from environment/naturalness — i.e., external interference is what turns ordinary intrinsic motion into excited, disturbed motion. This maps fairly directly onto §8 below (excitation), and the “orbital” component maps onto §6’s shell structure.

7.1 KD’s force/power distinction — and its indictment of physics. KD ch. 3.11 (“Force–Power (State–Motion)”) formalises a distinction the other texts leave implicit, and aims it at modern physics directly (our translations).

What MD says:

1. In state (*sthiti*), force (*bal*) is manifest; in motion (*gati*), power (*shakti*) is manifest (KD, p. 102).
2. Modern science has tried to recognise pressure as force, and motion/wave/flow as power (electrical waves, magnetic force, and so on); in doing so it denies the fact of mediation (*madhyasthīyakaran*) — all such study is based on excited (*āveshit*) motion alone, leaving unanswered where force and power “go” when something is in natural-state motion (KD, p. 102).
3. Every unit is force-endowed in state, and evidences power in motion (KD, p. 103).
4. Normal temperature means being in natural-state motion (KD, p. 75).

Our reading: *bal* (force) is what a unit *has* by virtue of being (its endowed, standing side — §4’s saturation-forcefulness); *shakti* (power) is what becomes evident when it is active or in motion (the expressed, relational side — §3’s relative energy). The pair maps onto *sthiti/gati* exactly as absolute/relative energy maps onto *satta/mutuality*. Claim 2 is the corpus’s most precise indictment of physics in physics’ own terms: its instruments and formalisms only ever register *excited* motion (disturbances, differences, §8 — SB’s “charged motion”), so its inventory of forces omits whatever holds in the *natural* state — which is where the *darshan* locates the mediative. Claim 4 closes the definitional circle from the heat side: the unexcited state is simultaneously the thermally ordinary one.

8. Charge and excitation (*āvesh*): generative/degenerative powers, the nucleus, and “hidden energy”

What MD says:

1. Activeness itself is recognised as generative, degenerative, and mediative powers; generative and degenerative powers are observed as the charge in mutuality, while the function of mediative activity is to normalise this charge — this mediative power is commonly known as the hidden energy (SB, p. 70).
2. A physicochemical atom’s positive or negative charge refers to the increase or decrease of orbital particles’ distance from the nucleus; positive and negative charge are only ever relative — “more” and “less” — never complete, and for that reason never stable, so mediative power works continuously to normalise them, because the goal of mediative activity is completeness (SB, p. 70).
3. (Same distance-from-nucleus definition of generative/degenerative excitation appears independently in AVD, p. 103 — see §6.2.)

4. In the regulated state there is no charge — this itself is the natural state; in the charged state, the possibility of decline naturally remains inherent, and decline is itself the evidence of charge (SB, p. 79).
5. Degenerative charge is always a manifestation of a problem (SB, p. 88).
6. An object that has become more excited is always heated — its temperature rises (KD, p. 74; cf. SB p. 252's heat = excitation, §9).
7. An atom must already be excited before it can shed a particle (KD, p. 57; §6.4b).
8. Pressure (*dabāv*) is the compulsion received for generative-degenerative (*sam-viṣam*) excitation, as distinct from natural-state motion (KD, p. 118) — folding SB p. 49's pressure definition (§2) explicitly into excitation vocabulary.
9. If the distance of an orbital particle from the nucleus increases beyond the definite good distance, the nucleus's mediating activity applies its force and draws the particle back in; if a particle begins to approach the nucleus too closely, the nucleus likewise applies its force and fixes it back at the good distance — this two-way restoring action is the atom's control activity (KD, p. 103).
10. Every excitation works toward natural-state motion; natural-state motion works toward development progression; development progression works toward evidencing development; development works toward evidencing awakening progression; and awakening progression works toward evidencing the continuity of awakening (KD, p. 77).

Terminology note: the Hindi root *āvesh/āveshit* is used **contextually** in the texts, and the translation must follow the context. **(a) The ± sense:** where the text speaks of the positive–negative (धन-ऋणात्मक) imbalance of atoms — defined by orbital particles' distance from the nucleus (claim 2; SB p. 250: “possible to measure positive and negative charges”) — *āvesh* is **charge**, in the electric-charge sense. **(b) The state sense:** where the text contrasts a unit's neutral/natural state with a disturbed one — *āveshit gati* vs *swabhav gati* (§7), an overfull atom's condition before expelling particles (SB pp. 72, 99, 259), a heated body's internal condition (SB p. 252) — *āveshit* is **excited**, and *āvesh* **excitation**. SB's published English uses “charge/charged” for both senses, but glosses the state sense itself at p. 257: “entities are in a natural state or in an excited (charged) state.” MD-Mapping encodes both readings (आवेश = “agitation, charge, excitement”; आवेशित गति = “agitated state, excited state”), and its Hindi definition of *āvesh* is explicitly value-laden — “deluded conduct-tendency; interference, motion toward decline.” This note and the KD-Karm-Darshan-English translations apply the contextual rule in their own voice; verbatim SB quotations elsewhere in this note keep the published wording. This settles, from the vocabulary side, what this section's reading argues from the content side: “charge” in the darshan names an imbalance-state, not an electrical quantity — and MD-Mapping's “motion toward decline” definition is precisely claim 4's “in the charged state, the possibility of decline naturally remains inherent.”

Our reading: the two senses of *āvesh* are systematically linked in SB's doctrine. In the $\pm$ sense, **charge** names the atom's generative/degenerative imbalance — generative power in excess of degenerative, or vice versa, measured by orbital distance from the mediative reference point at the nucleus (claim 2). Positive and negative charge are explicitly called *relative* — a matter of “more or less” around a mean, never an absolute or stable quantity — and mediative force's job is continuously pulling any excess back toward balance (claim 9 gives this a fully two-directional dynamical form — a restoring influence toward a set-point separation, against displacement in *either* direction; see §14.3 for what this does to the mediative-as-primitive claim). An atom carrying such an imbalance is *thereby* in the **excited state** — the state sense: the natural, unexcited state is definitionally the regulated (charge-free) one, and being excited is already partway toward decline. So the ontology runs: $\pm$ charge (imbalance) $\rightarrow$ excited state (disturbed condition) $\rightarrow$ decline, unless the mediative normalises it. This is a value-laden physics: excitation is functionally close to what SB elsewhere calls “problem,” and its absence is close to what it calls “resolution” (*samadhan*) — consistent with the book's title. Claim 10 embeds excitation in a universal directional chain; de-excitation is thus not merely local relaxation but the first step of a cosmic monotone — the darshan's arrow of change (taken up in §14.3).

9. Heat, magnetism, electricity, radiation

What MD says:

1. Radiance and imaging are not forms of motion or pressure; nevertheless, the intrinsic properties inherent in every entity exert influence on one another — heat, magnetism, electricity, and weight influence one another, and this field of influence is itself part of their grandeur (SB, p. 252).
2. Heat, magnetism, and electrical effects are projected on this same basis; fundamentally, heat equals the charge of a unit — the greater the internal charge within a planetary body, the more it tends to transform into atomic species of progressively lower numerical constitution (SB, p. 252).
3. Biological cells' mutual coordination and harmony is manifested through pulsation — contraction and expansion operating respectively as pressure and wave — and the entire biological-order unit is thereby proven to be electrically receptive (SB, p. 83).
4. Applying magnetic force to an atom generates a charge within it, leading to disintegration of its components; this reveals that the disintegrating force — electromagnetic force — is external to the atom itself, interacting with and altering its natural state (SB, p. 128).
5. Humans came to recognise the natural phenomenon of electricity through natural correlation, imaginativeness, and freedom of action, having already identified magnetic elements from prior events, which established the tradition of studying magnetism (SB, p. 128).

Our reading: heat, magnetism, electricity, and weight are grouped together explicitly as one family — properties that “influence one another” through a “field of influence,” and are explicitly *not* forms of motion or pressure in themselves (they are effects of the underlying excitation/pressure dynamics, not identical to them). Heat is given a specific reductive definition: heat = excitation (claim 2). Claim 4’s critique of experimental particle physics is philosophical, not technical: it reads magnetic/electromagnetic manipulation of atoms as an *externally imposed* distortion, and contrasts this with the darshan’s preferred method of studying wholes without destroying them (a recurring methodological theme — see also §6.4.2 of *The Ontology of Coexistence* on evidence standards).

9.1 The Sun, excess heat, and the “law of complementariness.” Directly following the heat = excitation passage, SB gives its one worked cosmological-scale application of the excitation doctrine.

What MD says:

1. A heated object (e.g. the Sun) reaches its natural state according to development progression; until such a state is established, the entire environment — including the Earth and other planets of comparatively lower temperature — distributes among itself the excess heat projected from the Sun, so that the Sun can become free from its excess heat through the law of complementariness (SB, p. 252).

Our reading: see §9.2’s synthesis below — the Sun’s cooling is read as approach to its natural state, not fuel consumption.

9.2 Radioactivity: “overfull” atoms and internally deployed heat. SB has an explicit account of radioactivity and nuclear explosion, entirely in the vocabulary of §6 (particle counts) and §8 (excitation and mediative regulation).

What MD says:

1. Atoms that have become overfull are radioactive by nature; radioactive substances are defined by intrinsic atomic heat or fire beginning to deploy internally — in an excited or interfered state, the pressure of this internalised heat may increase to the point of an explosive condition, and on this basis it appears reasonable to infer that in the Sun, the maximum number of overfull atoms have become charged, resulting in continuous nuclear explosions (SB, p. 259).
2. Human-triggered explosions occur because extracting substances that function beneath the Earth’s surface (per the law of complementariness) interferes with the nucleus’s naturally mediative activity — mediative force and power become incapable of balancing the charged components, leading to explosions (SB, p. 258).

Our reading: this closes a loop the earlier sections leave open. §6’s growth rule said more particles → more orbits → bigger nucleus; here we get the failure mode of that progression: an atom can become *overfull*, at which point the mediative force can no longer normalise its generative/degenerative excitations and the excess deploys internally as heat — radioactivity. The Sun is then read not as a fusion reactor but as a vast population of overfull, excited

atoms in continuous explosion, whose excess heat the cooler planets absorb (§9.1) until the Sun too reaches its natural state. Radioactive decay, solar output, and nuclear weapons are thus one phenomenon at three scales, and all three are excited states — i.e., in SB’s value-laden register (§8), problems awaiting resolution. Note the direction of explanation is the reverse of standard astrophysics: the Sun is cooling *toward* its natural state, not burning through fuel.

9.3 The critique extended: quantum mechanics, relativity, and fission. AVD broadens SB p. 128’s methodological complaint into a survey of physics’ own self-descriptions.

What MD says:

1. Specialisation has come to a purposeless dead end in all disciplines — quantum mechanics, for example, is facing the dead end of instability-uncertainty (AVD, pp. 63–64).
2. Relativity specialists use imagination to explain phenomena not completely understood so far — e.g. converting a magnetic field into electric current usable as work energy — and experiments are still being conducted to conclude whether waves should be categorised as made up of particles, with uncertainty still looming on the question (AVD, p. 64; the translator inserts a bracketed “From internet:” gloss on wave–particle duality at this point — a translator’s addition, not source text).
3. Nuclear fission — the splitting of the atom — was arrived at by a mathematical approach, discovered for making increasingly more powerful nuclear bombs (JV, p. 119).

Our reading: the corpus’s engagement with named modern physics is thus wider than SB p. 128 alone: Big Bang cosmology (§2.1), quantum mechanics and wave–particle duality, relativity, and fission are all addressed by name, and always with the same two-pronged objection — epistemic (uncertainty/instability as a *dead end*, not a finding) and moral (fragmentation-by-force as destruction, fission as bomb-making). Nowhere is the mathematics engaged on its own terms; the objection is to the method and its motive, consistent with SB’s own aphorism that mathematics is more than the eyes and less than understanding (SB, p. 87).

9.4 KD’s heat mechanics, waves, electricity, and celestial order. KD ch. 3.7 (“Heat and the Earth’s Balance”) and ch. 3.13 (“Pressure, Flow, Wave, Electromagnetic Force”) contain the corpus’s most worked-out phenomenal physics (all quotes our translation).

What MD says:

1. Heat only reflects (*parāvartit*) from a region of greater heat toward an adjacent region of lesser heat; heat spreads by molecules near the source becoming hot and expanding, passing heat molecule-to-molecule until everything arrives at normal temperature, which means being in natural-state motion (KD, p. 75).
2. Solids expand when heat is introduced, becoming liquid, then rarefied; heated iron keeps reflecting heat until it has cooled, distributing its heat among all associated

substances (KD, p. 74).

3. Waves occur differently by state of matter: in solid objects, waves are vibrational motion; in liquid, waves are the reaction produced by contact of air or objects; in rarefied substance, sound, heat, electricity, ray, and radiation are evidenced in wave form (KD, p. 118).
4. Sound arises from the friction or collision of one or more objects (two stones rubbed, metals struck, leaves in wind); word-waves travel by pressure imposed on the atmosphere's molecular groups, transmitted molecule to molecule across distance — radio and television waves are held to be word and sound propagated electricity-borne (KD, pp. 118–119).
5. An electric wave is the flow produced by the fragmentation (*vikhaṇḍan*) of magnetic current; the fragments into which magnetic current is broken are transmitted through a medium, and the whole of nature is found to be electricity-receptive, at minimum or maximum, with the degree settled by pressure, flow, and quantity (KD, p. 118).
6. Lightning is this electromagnetic wave occurring naturally; thunder is the sound of dense clouds nearing one another from their “good distance,” and when they connect with the Earth, electric flow runs of itself; the dynamo obtains light by the same fragmentation-of-magnetic-current method, rotated mechanically (KD, p. 119).
7. The Sun's heat is reflected equally in all directions; the Earth-facing side becomes hot, producing sunrise/sunset and the day's temperature curve. The Earth's rotation presents its faces to the Sun in turn; its revolution around the Sun, maintaining a “good distance” (*acchī dūrī*) along with all the planet-spheres of the solar array, follows an elliptical (*aṇḍākār*), not fully circular, path, whose chief intent is the Earth's self-organised tendency of seasonal change (*ṛtu parivartan*) (KD, p. 76).
8. The Sun is in the ultimate condition of decline, destined in the course of time to enter development progression (KD, p. 77).
9. Heat transfer generalises to astronomy: planet-orbs of maximum heat — such as the Sun's — have that heat absorbed into all facing planet-orbs in which less heat remains (KD, pp. 77–78).
10. The Earth's pole-to-pole magnetic current is what keeps the planet solid; if overheating deflected that current, the Earth would begin to scatter (KD, p. 81).

Our reading: three things stand out. First, the *direction* of KD's explanations: heat flows hot→cold not as statistics but as mutual “recognition–fulfilment” between substances (claim 2's melting iron “accepts” deployed heat — KD explicitly calls this acceptance *recognition*, extending §4's recognition doctrine into thermodynamics). Second, magnetism is treated as *prior to* electricity (claim 5: electricity = fragmented magnetic current) — the exact inverse of the electromagnetic hierarchy, but consistent with the corpus's decision to make “magnetic forcefulness” the fundamental endowment (§4). Third, the celestial mechanics (claim 7) is teleological: elliptical orbits and maintained “good distance” are not consequences of gravitation but expressions of order whose *purpose* is seasons, and ultimately the development-pro-

gression toward *jagriti* on Earth. Note the ellipse itself is correctly asserted — what differs from Kepler is the direction of explanation, the same pattern as §9.2's Sun. Claim 8 is the strongest single confirmation of §9.2's cooling-Sun reading; claim 9 is KD's version of SB p. 252's law of complementariness (§9.1); claim 10 gives magnetism, again, the structurally prior, binding role (cf. §4). One caution for the paper: KD ch. 3.12 shows *parāvartan–pratyāvartan* (reflection–projection, per the project's settled rendering) to be a fully general category — teaching/study, evidence-giving, and heat transfer are all its species — so ch. 3.7's thermal “reflection” (claim 1) should not be read as a specifically optical or thermodynamic technical term.

10. Mass and weight: the most concrete explanatory chain in the corpus

This is the best-developed physics-adjacent explanation in the primary texts, and it directly answers “how do atoms get weight/mass.”

10.1 The ground state is weightless. JV, in Q&A form, states this as an axiom.

What MD says:

1. Materialism holds that units, by displacing space, are mightier (heavier) than space; but in reality all units — Earth, Sun, solar systems, and galaxies — exist in a state of zero-attraction, the absolute state, meaning there is no weight in the coexistence of Omnipotence and units. Weight becomes evident only in the interaction of units — for example, in the mutuality of two atoms due to their tendency to form a molecule, or among molecules due to their tendency to form definite compositions. This manifestation of weight in the mutuality of units is what is commonly called the gravitational force (JV, pp. 149–150).

Our reading: weight is defined out of the ground state entirely and relocated to the tendency of units to combine — this is the axiom the rest of §10 builds on.

10.2 Three attraction-states, and a derivation chain for weight.

What MD says:

1. Every planet is in a state of zero attraction in space, its position determined by its motion; units in a state of positive-attraction are attracted to something, units in a state of negative-attraction attract something, and units in the zero-attraction state are independent (MVD, p. 45).
2. Weight depends on attraction; attraction depends on mutuality; mutuality depends on relative smallness or bigness; smallness or bigness depends on composition and relative energies; composition and relative energies depend on activity; activity depends on matter; and matter's development or decline depends on the right-use or misuse of relative energy (MVD, p. 45).

Our reading: the dependency chain in claim 2 traces weight backward through attraction → mutuality → relative size (*laghuta/guruta*, the same root that gives *gurutvakarshan*, the ordinary Hindi word for gravitation, so the etymology is doing real work here) → composition/relative-energy → activity → matter itself, terminating in “right-use or misuse of relative energy” — i.e., ultimately a normative rather than a purely mechanical vocabulary.

10.3 Atomic mass = particle count. (Same principle as §6’s orbit/nucleus growth rule, restated directly in terms of mass.)

What MD says:

1. The mass and lineage (or category) of atoms is determined by the number of their constituent particles; further development follows the principle of coexistence, with a two-particle atom being the simplest case (SB, p. 182).
2. The atom is established as the basic quantum: since molecules and molecular forms arise from atoms, understanding the atom as the fundamental unit of measurement provides the key to comprehending the quantity of all bodies (SB, p. 86).
3. More particles in an atom means more orbits also; as orbital particles increase, nucleus particles increase too — this is how weight-bondage appears in atoms (AVD, p. 185; see §6.1).

Our reading: atomic mass proper is simpler and more concrete than gravitation/weight (§10.1–10.2): it is just the count of constituent particles in an atom (nucleus + orbits, §6), which is also what fixes which “species” (element-analogue) the atom belongs to.

10.4 Sentience is explicitly freedom from weight.

What MD says:

1. Every sentient unit, i.e. *jeevan*, is free from weight-bondage and molecular-bondage (SB, p. 114).
2. A constitutionally complete atom is free from molecular and gravitational bonds, and this itself is its grandeur (JV, p. 150).

Our reading: the clean, repeated corollary of §§10.1–10.3 is that a *jeevan* atom (constitutionally complete, sentient) is explicitly exempted from both molecular bonding and gravitational/weight bonding — weight is presented as strictly a feature of insentient nature’s combinatorial tendency, one of the things a unit “graduates out of” at the sentience threshold.

10.5 KD: weight as evidencing coexistence, and “gravitation” renamed. KD ch. 3.11 gives the JV/MVD doctrine of §10.1–10.2 its most explicit form, including the one place in the corpus where the word *gurutvākarṣaṇ* (gravitation) is named and re-defined (our translations).

What MD says:

1. The whole of weight is the arrangement (*vinyās*) occurring in the course of evidencing coexistence with the Earth — weight is found to be animated by coexistence (KD, p.

103).

2. A solid object falling from above is the arrangement occurring with the Earth, in the effect of that object's evidencing its coexistence with the Earth; this arrangement is what has been given the name "gravitation" — in reality it is only the arrangement of a solid evidencing coexistence with a solid. From "gravitation" runs the conception of small and big, but seen as coexistence, each thing is complete together with its own environment, and there simply is no small or big (KD, p. 104).
3. This tendency is found to remain bounded (*maryādit*) up to this Earth's own environment; the Earth too is complete together with its own environment (KD, p. 104).
4. Molecules form when atoms join on the basis of the mid-particles' mutual attraction-force, operating through magnetic-force-endowment; molecules are thereby naturally weight-endowed, and because of this weight, one molecule tends to join another while staying at the definite good distance (KD, p. 103).

Our reading, pulled together (§10.1–10.5): weight/gravitation is, by explicit textual statement, not a property of matter in isolation and not something *satta* imparts. It is a name for a *tendency units display toward each other* — specifically, a tendency to combine into larger compositions (atoms→molecules→bodies) — that only becomes "evident" once more than one unit is in mutual relation. KD's contribution (claims 2–3) is to make the reversal of standard gravitation fully explicit: falling is not attraction by a larger mass but a body's *evidencing of coexistence* with the Earth; the small/big asymmetry that makes Newtonian gravitation asymmetric is dismissed as an artifact of the "relative method"; and — most testably — the tendency is *bounded to the Earth's own environment* rather than universal. That last clause puts KD in direct, statable conflict with universal gravitation (flagged in §15).

11. Indestructibility: nothing is created, nothing is annihilated

This theme is stated independently in all five texts and functions as the darshan's analogue of a conservation law — though what is conserved is *units*, not mass or energy as quantities.

What MD says:

1. Existence implies beingness and indestructibility (SB, p. 49).
2. Every unit is endowed with the dharma of existence, evidenced by the indestructibility of units — all that occurs is merely change and development; dharma means that from which one cannot be separated, and since existence is coexistence, this indestructibility is itself the ultimate dharma (SB, p. 50).
3. Even after fragmenting an atomic particle into countless parts, every fragment is still called "one" — demonstrating that annihilation or obliteration of anything is impossible; in existence there is no entity that "comes into being," nor anything that "ceases to be" (SB, p. 103).

4. From ancient times till now, all research, experiment, and conjecture on partitioning and division can be summarised as: nothing can be annihilated by progressive fragmentation (AVD, pp. 69–70).
5. An entity's decomposition does not mean annihilation of its respective constituents — existence neither decreases nor increases (AVD, p. 245; restated at p. 275: existence is therefore free from profit and loss, and thus indestructible).
6. However far we subdivide units (particles), it will only increase their number, without anything getting annihilated (AVD, pp. 247–248).
7. Every entity in its fundamental form is indestructible, no matter how many changes or results occur — this is the basis of the stability of the whole of existence (JVD, p. 82).
8. Physical entities continue to exist and are never annihilated; they only manifest as chemical entities, a process called chemical resonance (*rasaynik urmi*) (JV, p. 125).
9. There is no annihilation or absence of matter, *mun*, *vriddhi*, *chitta*, *buddhi*, or *atma* — they exist only in varying degrees of development and awakening (MVD, p. 299).

Our reading: the corpus's conservation principle is *countability*, not quantity: divide a unit as many times as you like and each fragment is still “one” — still bounded, still soaked in satta (cross-reference §4, claim 1: a unit cannot be annihilated because it cannot stop being soaked). What physics tracks as conservation of mass-energy the darshan replaces with a stronger, more ontological claim: composition and decomposition are the only real processes; creation *ex nihilo* and annihilation are not rare or difficult but strictly meaningless. This is the positive counterpart of §2.1's Big Bang rejection — no origin event is needed because “coming into being” is not something that happens. Note the asymmetry with physics: pair annihilation (particle + antiparticle → photons) would presumably count for the darshan as transformation of units, not annihilation, but no text addresses anything like antimatter, so the reconciliation is untested. Claim 9 extends the same principle across the sentient side — jeevan's faculties are as indestructible as particles, differing “only in degrees of development and awakening.”

12. Form — shape, volume, density — and the limits of measurement

The texts define every unit's *form* (*roop*) by a fixed three-dimensional triple, and then make an explicit claim about which of the darshan's quantities are measurable at all.

What MD says:

1. Form is understood in terms of shape, volume, and density (SB, p. 65; repeated as the first of the four dimensions of atomic bodies at SB p. 86, and as a definition at MVD, p. 50: form in all four orders is determined by differentiation of shape, volume, and density).

2. An object, in itself, is its shape, volume, and density — this is the very framework of a unit, defining its boundary; every unit is bounded on all sides, with Omnipotence seen on all sides of it (SB, p. 249).
3. Compositions made of atoms of a particular kind may vary in size, shape, volume, or density, but their essence remains constant — the intrinsic properties and nature of objects, expressed as their “essence,” do not increase or decrease despite variation in measurement, weight, or volume (SB, p. 119).
4. Shape, volume, and density can be measured, and it has become possible to measure positive and negative charges — but not all properties can be measured: mediative power, which underlies the natural state and mediativeness, has not been measurable (SB, p. 250).
5. An object’s volume is always more than what its shape projects to the eye, and its density even more than shape and volume — so whatever we see with our eyes, its entirety is not captured by the eyes (AVD, p. 199).
6. The shape, volume, and density of jeevan, a sentient unit, can be understood in the same context, since jeevan too is only an atom (AVD, p. 185).
7. However powerful the lens — showing something a billion or trillion times larger — instrument-evidence (*yantra pramāṇ*) cannot exceed the eyes: make the eye a billion times more effective, and only that much more meaning comes out (KD, p. 133).
8. Even mechanically-derived direction- and time-related activity remains related to the human at root; hence at the root of instrument-evidence, the human alone is the evidence (KD, p. 130); elsewhere KD presses the same point against science’s self-description — the instrument became treated as the principal evidence, with responsibility shifted from the human onto instruments, though with instruments no reasoning is possible at all, because what an instrument does, it keeps doing again and again (KD, p. 86).

Our reading: density is the third coordinate of unit-hood itself — form is shape-volume-density, and that triple is what bounds a unit off from the satta surrounding it (tying directly to §4’s “distance is satta”). Claim 4 then draws the corpus’s clearest line between what measurement can and cannot reach, and it maps exactly onto the §5/§8 force inventory: form dimensions — measurable; positive/negative (generative/degenerative) charges — measurable, and the texts credit physics with having measured them; **mediative power — not measurable**. This is the darshan’s own explanation of why the fifth force of §5’s table is absent from physics’ inventory: instrument-based method can only register what varies (charges, “more and less,” §8), and the mediative is definitionally the non-varying reference point of the variation. Claim 3’s essence-invariance (“ironness” constant under any change of size, weight, or volume) plays the role that intensive properties play in physics, but is extended into a metaphysical claim. Claim 5 adds an epistemic ordering — eyes capture part of shape, less of volume, less again of density — grounding the recurring argument that perception, and by extension instrumentation, systematically under-samples even measurable form.

Claim 7 supplies the missing premise connecting the two: instruments are *amplified eyes* and nothing more, so AVD's limits on the eye transfer to every microscope and detector — which is why, for the darshan, no instrument could ever register the mediative (claim 4) and understanding (*samajh*) rather than magnification is the required instrument.

13. Time (*kaal*) — brief cross-reference

Time is treated at length in the existing [Nature of Time](#) study; this note only pulls the handful of claims most relevant to the satta-unit physical-interaction question, for completeness.

What MD says:

1. Time (*kaal*) is the duration of an activity (MVD, p. 34).
2. KD opens with the identical definition, then critiques clock-time: time was originally recognised on the basis of one continuously recurring activity-cycle — the Earth's one rotational motion completing day-and-night, named “the day,” then divided into 24 parts, then 60, and so on. In this dividing, the activity itself was forgotten and time became treated as primary, an error from which further problems were born. Time is ever-present (*nitya vartmān*); the present, as reality, is coexistence itself, and it is to the present that the name *kaal* has been given (KD, p. 134).
3. Coexistence is *trikaalabadh* — a timeless truth valid across all three times (SB, p. 48).
4. Absolute energy differs from matter in that absolute energy is present in every time, place, and object, with no place or object ever found or proven devoid of it — whereas places devoid of matter do exist, though no time devoid of matter has been proven (MVD, p. 41).

Our reading: time is defined strictly as a property of unit-*activity* (*kriya*), exactly parallel to how §2–§3 above define pressure, force, and weight as unit-mutuality phenomena. *Satta* is *kriya-shunya* (actionless) and therefore, by the same logic, timeless — not because it exists “before” or “outside” time in a cosmological sense, but because time is definitionally absent wherever activity is absent. This is structurally the same move the texts make for force and weight: locate the physical-sounding predicate at the unit level, deny it of the ground, by definition rather than by discovery.

14. Synthesis: the structure of the darshan's physics

Read section by section, the corpus can look like a list of unconnected physics opinions. Read together, §2–§13 are one construction applied uniformly. This entire section is **our reading** — a synthesis built on the claims listed above, not a set of textual claims in its own right. It states the construction once (14.1), gives it a formal reading that organises it more tightly than paraphrase can (14.2), isolates the single claim in it that ordinary physics does not already hold in some form (14.3), and closes with the reference map (14.4).

14.1 The repeated move. Across pressure and cosmogony (§2), force (§3–4), atomic structure and states of matter (§6), charge/excitation (§8), heat/magnetism/electricity/radioactivity (§9), weight (§10), indestructibility (§11), form/density (§12), and time (§13), the texts do the same three things every time:

1. **Deny the predicate of *satta* directly and explicitly** — actionless, pressureless, wave-free, weightless (zero-attraction), timeless, non-transforming.
2. **Locate the predicate in unit-to-unit mutuality instead** — pressure from environmental influence; charge from generative/degenerative imbalance around a mediating centre; weight from units’ tendency to combine; time from activity’s duration.
3. **Name the bond between *satta* and units with force-like vocabulary anyway** — “forcefulness,” “magnetic forcefulness,” “energisation” — but frame it as an eternal endowment/unfolding, not an event, force-transmission, or efficient cause.

Steps 1 and 2 are repeated independently across SB, MVD, JV, and KD, and support a relational-dispositional axiom: **physical effects become manifest in mutuality between definite units; no such predicate belongs to the actionless ground.** This does not leave the units as predicate-free nodes. The texts attribute boundary, constitution, state, force-capacity, order, and *dharma* to the bearer, while MVD p. 40 defines heat, sound, electricity, pressure, waves, and fields as relative power that does not become apparent without unit mutuality. KD p. 102’s *bal/shakti* pair (§7.1) gives the same distinction dynamically: force is the property-endowed unit in state, power is the inseparable property evident in motion. Step 3 is the one place the system’s vocabulary pulls against itself (§15, first bullet) — though 14.2 suggests the tension is terminological rather than structural.

14.2 A formal reading: units as objects, mutuality as morphisms, *satta* as the unit object. *This subsection is our interpretive frame, not the texts’. It is included because it resolves two of §15’s open tensions and states the darshan’s position in vocabulary a physics-literate reader already has.*

Take units (*ikai*) as the objects of a category and mutualities as its morphisms. Objects retain typed structure — boundary, constitution, order, state, and capacities — while morphisms carry the actual reflection, relative influence, exchange, or composition through which those capacities become evident. A unit’s relational profile can therefore identify and test its kind (*jaati*, §6) without exhausting what the texts say the unit is. This is a narrower affinity with Machian, gauge-relational, and relational-quantum approaches than the claim that only relations exist: the comparison concerns interaction-indexed observables, not *svabhav*, *dharma*, development, or value.

Satta then maps not to any object in the category but to the **monoidal unit** — the “empty context” *I* with which every object composes without change ($A \cong I \otimes A \cong A \otimes I$). Every one of the texts’ descriptions checks against this role, clause by clause: *satta* “doesn’t prove to be a unit” (§2, claim 2 — precisely: *I* is not a system); it is formless (no internal structure); it is omnipresent and in-between (every composition of units factors through it — §4, claim 5’s “the definite distance between particles is formless existence”); and “saturation” is the *unitor*

— an isomorphism that is always already in place, structural rather than dynamical. On this reading the magnetic-force knot dissolves: the texts reach for force vocabulary because ordinary language has no word for a structural enabling condition, but the bond’s logical role is a unit law, not a dynamical arrow. “Being soaked itself is forcefulness” (§4, claim 1) becomes exactly right — the bond does nothing over and above being the condition of composition.

Three further pieces fall into place under the same reading:

- **Indestructibility (§11)** is the absence of a zero object: there are no morphisms from or to nothing. Composition and decomposition are rearrangements within the tensor structure; creation *ex nihilo* and annihilation are not rare or difficult — they are ungrammatical. This is why the texts treat “nothing can be annihilated” as a logical point about counting (“every fragment is still called ‘one’,” §11 claim 3), not an empirical discovery.
- **Time (§13)** is defined on morphisms — durations of activity — not as a background parameter. This is the process-first ordering that categorical approaches to physics also adopt: processes are primary, states derivative.
- **The Five Forces table (§5) and the universal constitution principle (§6, claim 3)** are claims that a structure-preserving map — a functor — exists between the insentient domain and the sentient one: same nucleus-and-orbits architecture, mapped across orders, not shared substance. This settles §15’s second bullet toward “analogical,” but with a precise sense of analogy that is stronger than loose metaphor: the claim is that the *structure* transfers exactly, whatever the entities are.

14.3 What the darshan asserts that physics does not: the mediative as primitive, and the development arrow. The relational axiom of 14.1–14.2 is congruent with live physics programs; it contributes an ontology, not new machinery. The mediative doctrine (§5, §8, §12) is different in kind, and it is the corpus’s one structural claim that ordinary physics *denies* rather than already believes. Physics treats equilibrium as emergent (a statistical fact about many interactions) and treats the reference point of any $\pm$ opposition as conventional (a choice of zero, a gauge choice). The darshan asserts the reference is ontologically primitive: a real third alongside every generative/degenerative pair, doing continuous regulatory work (§8, claim 2), and unmeasurable *by definition*, because measurement only registers variation and the mediative is the invariant of the variation (§12, claim 4). Stated physics-side, the general principle on offer is: **every observable two-valued variation presupposes a non-observable invariant that actively parameterises it**. The nearest analogues in physics are the connection/curvature distinction (curvature is measured; the connection itself is not) and renormalisation-group fixed points — but in both cases physics *derives* the invariant from the dynamics, whereas the darshan *posits* it as the more fundamental term and derives the dynamics (charge, excitation, decline, resolution) from distance to it.

KD both sharpens and strains this claim. It sharpens the *diagnosis*: physics misses the mediative because “all these studies are based on excited motion alone” (§7.1, claim 2) — mea-

surement registers variation, the mediative is what holds in the unvaried state. But §8’s claim 9 also gives the mediative an explicitly *dynamical* job description: a two-way restoring influence that pulls orbital particles back to a set-point “good distance” against displacement in either direction. That is, formally, a restoring force about an equilibrium separation — the kind of statement physics can model (a potential well) and does measure the consequences of (bound states, vibrational spectra). So KD makes the mediative doctrine less insulated than SB leaves it: a regulator that fixes an equilibrium distance has observable signatures even if the regulator itself is “not measurable.” The paper should note this as the corpus’s nearest approach to a falsifiable clause — and that physics would answer: the restoring behaviour around equilibrium separations is real and well-measured, and derives from the $\pm$ interactions themselves, needing no third primitive.

The full KD translation surfaces a *second* physics-denying claim, distinct from the mediative: **the direction of spontaneous change**. Physics’ second law says isolated systems drift toward equilibrium in the sense of disorder; the darshan asserts a universal drift with the opposite sign (§8, claim 10). The same monotone appears as §7’s rising vibrational motion *with rising stability*, §6’s development progression through species toward constitutional completeness, and §9.2’s Sun cooling *toward* its natural state — that last being simply this arrow applied to a star. So the corpus has two teleological primitives, and they compose: the mediative supplies the local restoring tendency (equilibration about set-points), and the development arrow supplies the global drift (successive set-points ascending toward completeness). A physics reader should be told plainly that this arrow runs against the entropic one, and that the darshan would answer the obvious objection (order increases locally only by exporting disorder) by denying the bookkeeping premise — for it, de-excitation is not export of disorder but approach to the natural state, which costs nothing.

Two liabilities, flagged for the paper: (i) the unmeasurability doctrine insulates the mediative claim from every empirical test — a physics reader will identify this immediately, and the paper should concede it rather than soften it (with the KD qualification above); (ii) §11’s conservation of *units* (countability) conflicts with particle creation and annihilation in quantum field theory unless the darshan’s “particles” sit below field quanta — a question the texts never face (no antimatter, no field-quantum vocabulary anywhere in the corpus).

14.4 Reference map. A rough map, for reference:

Physics vocabulary	Primary-text location	Level assigned
Pressure, attraction, repulsion	SB p. 49	Unit mutuality
Heat, sound, electricity, waves, fields (“relative energy”)	MVD p. 40	Unit mutuality

Physics vocabulary	Primary-text location	Level assigned
Magnetism / electromagnetic force	SB pp. 85, 128; KD pp. 54, 103, 118	Unit mutuality (named force); also used for the saturation bond itself (§4)
Electricity (“fragmentation of magnetic current”), waves, sound	KD pp. 118–119	Unit mutuality (medium-borne; all nature electricity-receptive)
Gravitational force / weight	JV pp. 149–150; MVD p. 45; KD pp. 103–104	Unit mutuality (tendency to combine; “evidencing coexistence with Earth” — KD bounds it to Earth’s environment)
Weak / strong interaction	SB p. 85	Unit mutuality (atomic-scale, table only — no further gloss found)
“Mediative force”	SB pp. 84–85; KD pp. 102–103	Unique to darshan; atomic-nucleus-level, regulates the other four (KD: two-way restoring toward “good distance”)
Atomic structure (nucleus + orbits, “kind”/element)	MVD pp. 42, 78, 149, 174, 317; AVD pp. 103, 185; KD pp. 53–57	Unit-internal (particle count and arrangement; KD: nucleus ≥ orbits, four-orbit equilibrium, 108 species acknowledged)
The constituent particle (<i>ansh</i>) itself	MVD pp. 89, 174, 317; SB ch. 4; KD pp. 54, 56–57	Sub-unit-internal (functional role by position; force-endowed by saturation; restless/incomplete in isolation — §6.6–6.9)
Charge / excitation (<i>āvesh</i> — context-dependent, §8)	SB pp. 70, 79, 88; AVD p. 103; KD pp. 57, 74, 118	Unit-internal: ± charge = generative/degenerative imbalance (orbit-to-nucleus distance); excited state = the disturbed condition it produces (KD: precondition of particle transfer)
Mass (atomic)	SB pp. 86, 182; AVD p. 185; KD p. 56	Unit-internal (particle count in nucleus + orbits; KD keys weight to nucleus count)
States of matter (solid, liquid, gaseous)	SB pp. 97, 142, 145; MVD pp. 41–42; JV p. 14	Unit composition (three states only; no plasma)
Radioactivity / nuclear explosion	SB pp. 258–259	Unit-internal (overfull atoms; mediative regulation overwhelmed)

Physics vocabulary	Primary-text location	Level assigned
Density (with shape, volume: form)	SB pp. 65, 86, 119, 249–250; MVD p. 50; AVD pp. 185, 199	Unit-internal (the boundary/framework of a unit)
Indestructibility (no creation, no annihilation)	SB pp. 49–50, 103; AVD pp. 69–70, 245, 247–248; JVD p. 82; JV p. 125; MVD p. 299	Dharma of every unit (conservation analogue — units, not quantities)
Cosmogony (Big Bang, rejected)	JV p. 149	Denied of satta (forceless ground); no origin event needed
Celestial mechanics (elliptical orbits, seasons, solar heat)	KD p. 76; SB pp. 252, 259	Unit mutuality (“good distance”; teleological — order for the sake of seasons/development)
Time	MVD p. 34; SB p. 48; KD p. 134	Unit activity (KD: clock-time a derived convention; “time became primary, the object forgotten”)
Development arrow (excitation → natural-state → development → awakening)	KD p. 77; SB pp. 53, 62; MVD p. 149	Universal directional drift of unit-states — opposite in sign to the thermodynamic arrow (§14.3)
Absolute / uniform energy	MVD pp. 40, 45; SB pp. 48–50	Satta itself — not a force, the ground of all the above
Forcefulness / “magnetic forcefulness” of saturation	SB pp. 57, 61–62, 79, 98	The satta–unit bond itself (endowment, not efficient causation)

15. Open interpretive tensions (flag for the paper, not yet resolved)

This section, like §14, is entirely **our reading** — flagged tensions and open questions, not textual claims.

- **“Magnetic force” vs. actionless doctrine.** SB literally calls the saturation bond’s forcefulness “magnetic force” (चुम्बकीय बल, p. 61) while also insisting *satta* is actionless and non-transforming. The texts do not reconcile these in the same passage. The most charitable reading (used in *The Ontology of Coexistence* §1.2) is that “magnetic force”

names a standing, eternal endowment rather than a transmitted quantity — but this is an interpretive choice, not something the text states outright. §14.2’s formal reading (saturation as unit law, not dynamical arrow) makes the choice principled rather than merely charitable. Worth a dedicated footnote or subsection wherever this note’s material gets folded into a published study.

- **Five Forces table — descriptive or analogical?** SB p. 85 does not say whether it is claiming physics is incomplete without a fifth (“mediative”) force, or simply using the four-force inventory as a structural analogy for jeevan’s five faculties. The strong/weak interaction rows get no independent gloss anywhere else in the corpus (only the table itself) — thin evidentiary support compared to electromagnetic/gravitational, which are discussed extensively elsewhere. §14.2’s functor reading favours the analogical horn, with a precise sense of “analogy” (structure transfers exactly; substance does not).
- **Āvesh as evaluative vocabulary.** SB ties charge/excitation definitionally close to “problem” and its absence to “resolution” — useful for the book’s thesis (Resolution-Centred Materialism) but worth flagging as a place where physics vocabulary is doing double duty as ethical/soteriological vocabulary; MD-Mapping’s own definition of *āvesh* (“deluded conduct-tendency; motion toward decline”) builds the evaluation into the word. The contextual rule (§8: $\pm$ imbalance = charge; disturbed condition = excited state) resolves most instances, but a reader could still ask whether *āvesh* is one concept with two faces or two concepts sharing one word — the texts treat the $\pm$ imbalance and the disturbed state as one phenomenon, which is itself a substantive claim.
- **No subatomic-particle vocabulary, no periodic table, no atomic numbers.** Electron, proton, neutron do not appear in SB, MVD, JV, AVD, JVD, or KD. “Particles” (*ansh*) are counted generically, and no named element (hydrogen, iron, gold, etc.) is ever given a specific particle count — only a growth rule (§6) now supplemented by KD’s structural constraints (§6.4a: minimum 1+1, nucleus $\geq$ orbits, four-orbit equilibrium) and KD p. 55’s acknowledgement of the “108 species” inventory as a human claim. Any comparison to modern atomic physics needs to keep the remaining gap explicit: the darshan has the *shape* of a shell model, and now some of its *inequalities*, but still no per-element numbers (no analogue of “carbon = 6”). The one place a specific, named, fixed structure does appear is *jeevan* itself — exactly four orbits plus nucleus, ten activities, psychologically named rather than numbered (§6.4) — which is a special case, not the general insentient-atom model. (Curiously, KD’s nucleus $\geq$ orbits inequality *holds* under the natural modern mapping: if orbital *ansh* = electrons and *madhyansh* = nucleons, then “at least as many in the middle as in the orbits, possibly more” is exactly $A \geq Z$ — true of every neutral atom, with equality only at hydrogen-1, which is also KD’s stated minimum atom (1+1). Whether this is insight or coincidence cannot be settled from the texts, since *ansh* is never identified with any named particle — but it is the corpus’s single most striking numerical agreement with standard atomic physics and deserves a paragraph in the paper.)

- **Is particle-level restlessness (§6.8) universal, or specific to sub-threshold particles?** KD’s “whirling in the void” passage (p. 56) is stated for atoms still under development; whether the same claim would apply to, say, a *buddhi*-particle hypothetically separated from an already-completed *jeevan* atom is not addressed anywhere in the corpus.
- **Does the nucleus/orbit functional split plus subtlety gradient (§6.6) generalise beyond *jeevan*?** MVD names the gradient (*mun*→*atma*) only for the sentient atom; whether an ordinary (insentient) atom’s nucleus particles are, in some lesser sense, also “more developed” than its orbital particles, or whether the gradient reading is *jeevan*-specific, is not settled by the texts consulted here.
- **“Overfull” is undefined.** SB p. 259 rests radioactivity (and the entire solar model of §9.2) on atoms being “overfull,” but no text says overfull relative to *what* — there is no stated maximum particle count per orbit or per nucleus (consistent with §6’s missing numbers, but here the gap is doing explanatory work). Also unreconciled: §9.2’s Sun-as-cooling-toward-natural-state reverses the direction of standard stellar physics, and the texts never engage fusion as an energy source at all.
- **Jeevan’s four orbits vs. insentient atoms’ open-ended orbit count — now partially answered by KD.** MVD p. 78 allows ordinary atoms “first, second, third, fourth, and so on” orbits (unspecified upper bound) but fixes *jeevan* at exactly four, without explanation. KD p. 56 (§6.4a) supplies what looks like the missing law: orbit counts above four tend to decrease and below four tend to increase — four is the equilibrium of the orbit-count dynamics. On that reading, constitutional completeness is the *stable fixed point* of atomic development, and *jeevan*’s four orbits are not an arbitrary cap but the attractor every developing atom moves toward — which meshes exactly with §6.4b’s *prasthāpan–visthāpan* (the particle-transfer process that moves atoms through species) and with MVD p. 78’s open-ended counts for atoms still en route. Still open: *why* four is the equilibrium (KD asserts the tendency, not its ground), and whether the §1.6.1 (*gathanpurnata*) material glosses completeness as specifically the four-orbit configuration.
- **KD’s gravitation is local, not universal.** KD p. 104 bounds the falling-tendency “up to this Earth’s environment,” with each body “complete with its own environment” (§10.5). Taken at face value this denies universal gravitation — yet §9.4 (KD p. 76) has the planets maintaining “good distance” around the Sun, which universal gravitation explains and a strictly Earth-bounded tendency does not. The texts do not connect the two passages. Options for the paper: (a) “good distance” order among planets is a separate, non-gravitational principle for KD (consistent with JV’s zero-attraction doctrine, §10.1); (b) the boundedness claim concerns only the *falling* arrangement, not inter-body order. The corpus underdetermines the choice.

Method note (for reproducibility)

Primary texts were converted to plain text with `pdftotext -layout`, and page numbers were cross-checked two ways: (1) internal footer markers (“page-NN”) embedded in the MVD and JVD PDFs, present roughly every page; (2) for SB and JV, where footers were not machine-readable, page indices were calibrated against page citations already established and used consistently in *The Ontology of Coexistence* (e.g. “SB, p. 48,” “SB, p. 62,” “MVD, p. 40”) by locating the matching English sentence and checking the offset was constant (confirmed: PDF page-splitting index + 1 = printed page number, holding across more than five independent check-points in SB). The claims above are paraphrased from the English portion of the bilingual PDFs; Hindi originals were consulted for the technical terms glossed in brackets. Before quoting any claim verbatim in a catalog Study, re-verify the exact wording against the source PDF via `Scripts/_quote_tool.py verify --study <Slug>` — the paraphrase here is for readability and is not itself citable wording.

KD (Manav Karm Darshan, v5) requires a separate protocol. The PDF is Hindi-only (no English translation exists in the References), and its embedded font’s character map is corrupted, so machine-extracted text is systematically garbled (e.g. त/स/ि cycle, श↔र) — the text layer was used only to *locate* candidate pages by cipher-adjusted term search, never to transcribe. All KD passages cited here were read visually from rendered page images and translated by us. **The full working English translation (front matter + chapters 1–3) is in** `References/Madhyasth-Darshan/KD-Karm-Darshan-English/KD-Karm-Darshan-English.md`, with `[p. NN]` printed-page markers, `_page-images/` Hindi page renders, and term decisions in `KD-Glossary-Additions.md` + `KD-Translation-Glossary.xlsx`; inline KD claims in this note should be read against that file, which supersedes them where wording differs, and all remain **working translations pending a published English edition**. Page numbers cited are the printed page numbers in the running headers (calibration: printed page = PDF page – 25, constant across all pages checked). Two fixed terminology conventions: *āvesh/āveshit* is translated **contextually** — “charge” where the text concerns the positive/negative (generative/degenerative) imbalance of atoms, “excited/excitation” where it concerns a unit’s disturbed state as against its neutral/natural state (rule stated in full in §8’s terminology note; SB’s published English uses “charge” for both and glosses the state sense “excited (charged)” at p. 257; MD-Mapping carries both readings); and *śram* = **effort** (MD-Mapping’s notes: “We are using ‘effort’ for shram in the sense of shram-gati-parinam”), with “labour” reserved for economic compounds (श्रम मूल्य = labour value). Chapter 3 (printed pp. 53–153) is fully translated in the combined file; ch. 3.12 (reflection–projection) proved on full translation to be chiefly epistemological — teaching/study as reflection–projection — with heat transfer as one species of the general category (noted in §9.4).